

SMART INDIA HACKATHON 2026



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HACKATHON
2026**

TITLE PAGE

- Problem Statement ID –26106
- Problem Statement Title-AI Powered Email Threat Detection , GeoLocation and Forensic Intelligence Platform
- Theme- Block Chain And Cyber Security
- PS Category- Softwar
- Team ID-
- Team Name – Stack Overflowed



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❖ Proposed Solution

- Detailed explanation of the proposed solution

Unified Intelligence Engine: An automated platform that ingests raw email headers, body content, and attachments to detect complex cyber threats in real time.

Core Capabilities: Combines natural language processing (NLP) for phishing detection, metadata parsing for IP/mail route extraction, and IP geolocation mapping to trace attacker origin points.

Forensic Analytics: Generates structured forensic timeline reports detailing mail transport history (Received headers), SPF/DKIM/DMARC validation failures, and hidden network hops

How It Addresses The Problem

Eliminates Manual Header Analysis: Reduces forensic investigation time from hours to seconds by automating email header decompilation and route visualization.

Detects Spoofing & Phishing: Identifies spoofed domains and zero-day phishing links using machine learning before users interact with malicious content.

Actionable Threat Tracing: Maps server hop routes directly onto a global geographic map, highlighting anomalous origin points (e.g., internal email routed via external unexpected IP).

Innovation and Uniqueness

Integrated Forensic Chain: Merges linguistic threat analysis with physical network topology tracing in a single view.

AI-Driven Spoofing Engine: Uses fine-tuned NLP and anomaly detection models to flag subtle domain lookalikes (typosquatting) and header inconsistencies.

Automated Evidence Export: Generates court-ready/incident-response PDF logs containing cryptographic hashes (SHA-256) of raw email files.

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TECHNICAL APPROACH



- Technologies to be used

Frontend: React.js, Leaflet.js / Mapbox (for real-time geolocation mapping), Chart.js

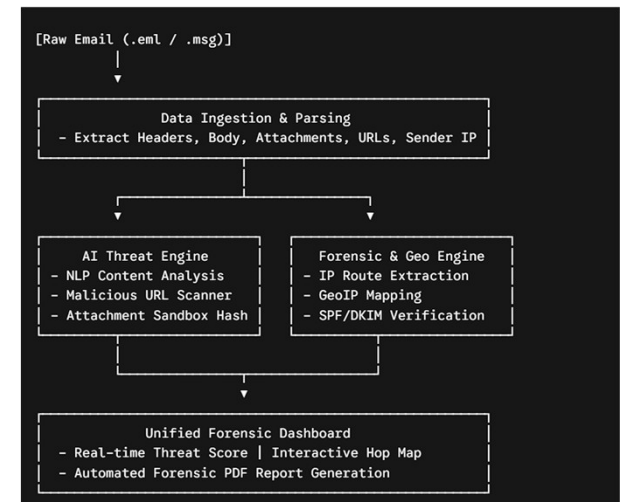
Backend: Python (FastAPI / Django Framework)

AI/ML: PyTorch / Transformers (BERT/Roberta for phishing text classification), Scikit-learn (anomaly detection)

Forensics & Intelligence: MaxMind GeoIP2 / IPinfo API, dnspython (SPF/DKIM/DMARC checks), Python email & BeautifulSoup libraries

Database: PostgreSQL (structured metadata storage), Redis (caching threat intelligence feeds)

- Methodology and process for implementation (Flow Charts/Images/ working prototype)



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FEASIBILITY AND VIABILITY



1. Feasibility Analysis

Technical Feasibility: Highly feasible using open-source Python forensic packages, pretrained Transformer models for NLP, and existing GeoIP APIs.

Scalability: Microservices architecture allows independent scaling of AI inference pipelines and heavy mail header parsing.

2. Potential Challenges and Risks

Evasion via VPNs/Proxies: Attackers often route emails through anonymizing proxies or VPN nodes, obscuring real IP origins.

Parsing Edge Cases: Non-standard or malformed email headers across different mail transfers (MTA implementations).

False Positives: Legitimate promotional or third-party marketing emails failing SPF/DKIM checks due to misconfigurations.

3. Mitigation Strategies

Proxy/Tor Detection Integration: Cross-reference extracted IPs against known Tor exit node lists, public VPN subnets, and threat feeds.

Standardized RFC Parsing: Implement resilient fallback parsers adhering strictly to RFC 5322 and RFC 2822 specifications.

Hybrid Scoring System: Weighted risk scoring combining ML intent analysis, header integrity checks, and IP reputation rather than relying on a single metric.

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IMPACT AND BENEFITS



1. Potential Impact on Target Audience

Security Operations Centers (SOC): Accelerates triage time for reported phishing emails and security incidents.

Law Enforcement & Forensics Teams: Provides structured, chronological evidence trails for cybercrime investigations.

Enterprise IT Admins: Offers clear visibility into domain spoofing attempts targeting corporate communications.

2. Key Benefits

Economic: Minimizes financial losses associated with Business Email Compromise (BEC) and ransomware deployments delivered via phishing.

Operational: Automates repetitive tier-1 SOC tasks, freeing cybersecurity personnel for high-priority threats.

Social / National Security: Enhances national cyber resilience by equipping Indian organizations with low-cost, high-efficiency forensic tools.

RESEARCH AND REFERENCES



Analysing E-Mail Headers for Forensic Investigation (Banday, 2011)

Focus: Detailed header deconstruction (Received fields, Return-Path, X-Headers) and forensic investigation techniques to identify spoofed original senders and hop routes.

Link: [Download PDF from ERAU Digital Commons](#)

Analysis of Email Header for Forensics Purpose (Guo et al., 2013)

Focus: Technical breakdown of Mail Transfer Agent (MTA) hops, header field integrity validation, and origin tracking for legal evidence.

Link: [View Document on IEEE Xplore / ResearchGate](#)

Phishing Attacks Detection: A Machine Learning-Based Approach (Salahdine et al., 2021)

Focus: Machine learning classifiers (SVM, Logistic Regression, Neural Networks) evaluated on real-world phishing datasets and email metadata.

Link: [Read PDF on arXiv](#)

Revisiting Email Spoofing Attacks (arXiv:1801.00853)

Focus: Empirical study analyzing authentication protocols (SPF, DKIM, DMARC) and modern techniques used to forged email transport routes.

Link: [Read on arXiv](#)

Context-Aware Phishing Email Detection Using Machine Learning (arXiv:2603.27326)

Focus: NLP-driven feature extraction pipelines designed to detect context-aware phishing cues and social engineering intent within raw email content.

Link: [Read PDF on arXiv](#)